

Quiz 2 - Pumping Lemma

Taken on: September 14, 2024

Total: 10 points

Question 1

[5 points]

Prove that the language $A_1 = \{www \mid w \in \{0,1\}^*\}$ is not a regular language.

Understanding about the language

Strings that are valid in this language have a specific part w which repeats three times. For example, if $w = 01$ then a string would be 010101 which is accepted in this language. Whereas, the string 0110 would not be accepted in this language as we are unable to identify w which repeats 3 times.

Solution

To prove that A_1 is a non-regular language, we will perform a proof by contradiction. Assuming A_1 is a regular language, we assume the pumping lemma with pumping length p is valid.

Assuming the string

$$s = 01^p 01^p 01^p$$

which has $w = 01^p$, this string s is in the language A_1 and $|s| = 3p + 3$ which is greater than the pumping length p , hence string s is valid.

Condition $|s| \geq p$ is proved.

The three conditions of the pumping lemma are:

Condition 1: $xy^i z \in A_1$ where $i \geq 0$.

Condition 2: $|y| > 0$.

Condition 3: $|xy| \leq p$.

To split this string s into 3 parts x, y, z , we can perform the split,

$$x \rightarrow 0, y \rightarrow 1, z \rightarrow 1^{p-1} 01^p 01^p$$

Condition 2, $|y| > 0$ is $|1| = 1 > 0$ thus validated.

Condition 3, $|xy| \leq p$ is $|01| = 2 \leq p$ thus validated.

Pumping down:

To perform condition 1, let us assume

$$i = 0 \text{ where the string will be } xz \text{ which is } 01^{p-1} 01^p 01^p$$

The pumping lemma states that this pumped-down string should be in the language A_1 but after pumping down, we can see that this string is not in the language, as there is no repeating w .

Pumping up:

To perform condition 1, let us assume

$$i = 2 \text{ where the string will be } xyyz \text{ which is } 0111^{p-1} 01^p 01^p \text{ which is } 01^{p+1} 01^p 01^p$$

The pumping lemma states that this pumped-up string should be in the language A_1 but after pumping up, we can see that this string is not in the language.

Thus pumping lemma has been violated and therefore we have proof by contradiction that the language A_1 is not a regular language.

Marking Criteria

- + 1 for correctly defining a string s which is in the language and is in terms of p
- + 1 for validating condition 2 that $|y| > 0$
- + 1 for validating condition 3 that $|xy| \leq p$
- + 1 for correctly pumping up/down and showing the string after pumping
- + 1 for a concluding statement and stating that this pumped string is not in the language therefore this language is a non-regular language and a proof by contradiction

Question 2

[5 points]

Prove that the language $A_2 = \{0^n 1^m 0^n \mid m \neq n\}$ is not a regular language.

Understanding about the language

Strings that are valid in this language have equal 0s on each side and more or less 1s in the middle. For example, strings 0011100 and 00100 are in the language A_2 but the string 001100 is not.

Solution

To prove that A_2 is a non-regular language, we will perform a proof by contradiction. Assuming A_2 is a regular language, we assume the pumping lemma with pumping length p is valid.

Assuming the string

$$s = 0^p 1^{p+1} 0^p$$

this string s is in the language A_2 and $|s| = 3p + 1$ which is greater than the pumping length p , hence string s is valid.

Condition $|s| \geq p$ is proved.

The three conditions of the pumping lemma are:

Condition 1: $xy^i z \in A_2$ where $i \geq 0$.

Condition 2: $|y| > 0$.

Condition 3: $|xy| \leq p$.

To split this string s into 3 parts x, y, z , we can perform the split,

$$x \rightarrow 0, y \rightarrow 0, z \rightarrow 0^{p-2} 1^{p+1} 0^p$$

Condition 2, $|y| > 0$ is $|0| = 1 > 0$ thus validated.

Condition 3, $|xy| \leq p$ is $|00| = 2 \leq p$ thus validated.

Pumping down:

To perform condition 1, let us assume

$$i = 0 \text{ where the string will be } xz \text{ which is } 00^{p-2} 1^{p+1} 0^p \text{ which is } 0^{p-1} 1^{p+1} 0^p$$

The pumping lemma states that this pumped-down string should be in the language A_2 but after pumping down, we can see that this string is not in the language, as the first and last 0s are not equal.

Pumping up:

To perform condition 1, let us assume

$$i = 2 \text{ where the string will be } xyyz \text{ which is } 0000^{p-2} 1^{p+1} 0^p \text{ which is } 0^{p+1} 1^{p+1} 0^p$$

The pumping lemma states that this pumped-up string should be in the language A_2 but after pumping up, we can see that this string is not in the language, as now $m = n$ and the first and last 0s are not equal.

Thus pumping lemma has been violated and therefore we have proof by contradiction that the language A_2 is not a regular language.

Marking Criteria

- + 1 for correctly defining a string s which is in the language and is in terms of p
- + 1 for validating condition 2 that $|y| > 0$
- + 1 for validating condition 3 that $|xy| \leq p$
- + 1 for correctly pumping up/down and showing the string after pumping
- + 1 for a concluding statement and stating that this pumped string is not in the language therefore this language is a non-regular language and a proof by contradiction